

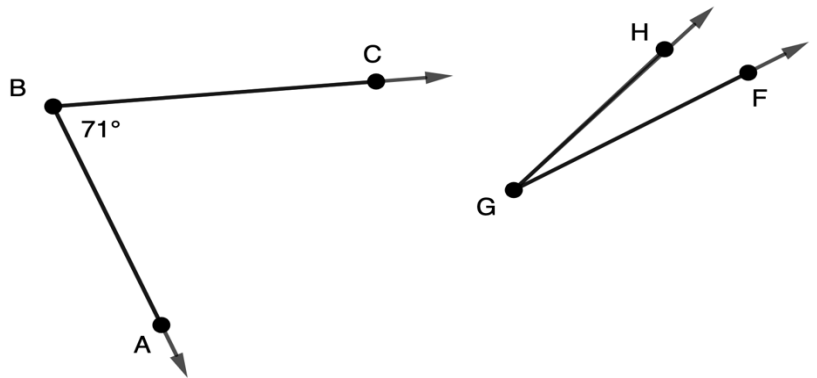
Geometry
Unit 2
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

Complete the two-column proof.

Given: $\angle ABC$ and $\angle HGF$ are complementary; $m\angle ABC = 71^\circ$

Prove: $m\angle HGF = 19^\circ$



	Statement	Reason
1.	$\angle ABC$ and $\angle HGF$ are complementary	Given
2.	$m\angle ABC + m\angle HGF = 90^\circ$	Definition of complementary angles
3.	$m\angle ABC = 71^\circ$	Given
4.	$71^\circ + m\angle HGF = 90^\circ$	Substitution Property
5.	$m\angle HGF = 90^\circ - 71^\circ$ $m\angle HGF = 19^\circ$	Subtraction Property of Equality

Complete the paragraph proof to answer questions 6-10.

Given: $\angle 1$ and $\angle 2$ are complementary and $\angle 2$ and $\angle 3$ are complementary.

Prove: $\angle 1 \cong \angle 3$

It is given that 6. $\angle 1$ and $\angle 2$ are complementary and $\angle 2$ and $\angle 3$ are

complementary. By the definition of complementary angles, 7. $m\angle 1 + m\angle 2 = 90^\circ$

and $m\angle 2 + m\angle 3 = 90^\circ$. By the 8. transitive property, $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$.

Then, $m\angle 1 =$ 9. $m\angle 3$ by the subtraction property. Therefore, $\angle 1 \cong \angle 3$ by the

10. definition of congruence